



Chapter 8: Sights & Sounds (Media)

I. Image Formats: Quality vs. Speed

- **Core Principle:** The image format you choose has a huge impact on your site's quality and _____ (how fast it loads).
- **Definition:** **Lossy** compression _____ some data to get a smaller file size. **Lossless** compression _____ all data, never losing quality.

Format	Pros	Cons	Best Use Case
JPG (JPEG)	Handles millions of colors; compresses to very _____ file sizes.	It is _____ compression (loses data).	_____ (real-world photographs).
PNG	It is _____ compression; supports _____ backgrounds.	File sizes can be much _____ than JPGs.	Logos, icons, and graphics that need to "float."
GIF	Supports _____ .	Terrible quality; limited to _____ colors.	Short, looping animations (e.g., memes, loading spinners).
SVG	It is _____ (made of code); _____ scalable (looks sharp everywhere).	Not suitable for _____ .	_____ and simple graphics.

Deep Dive Discussion

1. **Logo Choice:** If your company logo is a simple two-color design, why should it **always** be an SVG file instead of a PNG or JPG?

2. **Lossy Risk:** You have a beautiful JPG photo on your site. Why should you avoid re-saving that JPG file 100 times during the design process?

Application Activity

1. **Image Audit:** You are building a landing page. For each asset below, state the **best image format** to use and why.
 - **Asset A:** A photograph of your team at a company retreat.
 - Format: _____. Reason: _____
 - **Asset B:** A simple, three-color icon that appears next to every link.
 - Format: _____. Reason: _____

II. Embedding Video and Audio

The Easy Way (YouTube/Vimeo)

- **Tag Used:** An _____ tag creates a "window" to another website inside your own.
- **Benefit:** This approach is easier because it offloads the _____ drain of hosting and delivering large video files to the user.

The Hard Way (Native HTML5 Media)

- **Purpose:** Used when you **must** host the file yourself (e.g., background video loops, proprietary audio).

Media Tag	Required Attribute	Key Element/Content
<video>	_____ (adds the play/pause button, volume, etc.)	Uses multiple _____ tags (e.g., .mp4, .webm) to provide browser choices.
<audio>	_____	Uses multiple source tags (e.g., .mp3, .ogg) for maximum _____. _____.

Deep Dive Discussion

1. **Source Fallback:** Why do the <video> and <audio> tags use multiple <source> elements

(like both .mp4 and .webm or .mp3 and .ogg)?

2. **Bandwidth:** You are embedding a 4K video of a rocket launch on your site. Why is it a bad idea to use the native `<video>` tag and host the file on your cheap shared hosting server?

III. Displaying Tabular Data

- **Strict Rule:** Tables are **ONLY** for displaying _____ (like a spreadsheet).
- **The Forbidden Rule:** **NEVER** use tables for _____ (use Flexbox or Grid instead!).

HTML Tag	Purpose	Description
<table>	The main _____	The container for the entire table.
<tr>	_____	Defines a single row in the table.
<th>	_____	A header cell (content is usually bold/centered).
<td>	_____	A regular content cell.

Application Activity

1. **Table Structure:** The goal is to display a short list of three console types and their max resolution. Write the **four** necessary HTML tags in the correct order to create the **header row** of this table (you do not need the table content).

- **HTML Code:**